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Spatial Aware Peak Ranking in Mass Spectrometry Imaging

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***Abstract*—Mass spectrometry imaging (MSI) records thousands of mass-to-charge (m/z) channels at every pixel of a tissue section, so analysis usually begins by selecting a few informative peaks. Learned methods such as msiPL rank peaks from an unsupervised model, and S^{3}PL adds spatial information through 3D convolution. We ask whether local neighbourhood context improves learned peak ranking and where in the pipeline it matters. Variational autoencoders with and without neighbourhood input were segmented with a Gaussian mixture model. Peaks were ranked either by Integrated Gradients (IG) through the model or by correlating each ion image with the segment posterior maps. On 24 sections from three public datasets, at matched peak budgets and under predeclared decision rules, correlation outperformed IG in 140 of 144 paired comparisons across three seeds, a difference that was statistically supported for every dataset and model. In these tests, neighbourhood input to the encoder gave no consistent, spatially specific gain, while spatially aware ranking statistics were the strongest label-free rankings on two of the three datasets and within 0.01 of the best on the third.**

***Index Terms*—mass spectrometry imaging, peak selection, variational autoencoder, spatial context, Integrated Gradients, reproducibility**

I. INTRODUCTION

Mass spectrometry imaging (MSI) records a full mass spectrum at every pixel of a tissue section, so the spatial distribution of many molecules can be studied at once without choosing targets in advance [Caprioli et al. 1997; Schwamborn and Caprioli 2010; Buchberger et al. 2018]. The resulting data are large and high-dimensional. The collections used here contain between about a thousand and several thousand pixels per section, and between 1,481 and 85,062 mass-to-charge (m/z) bins per spectrum. Most analyses therefore begin by selecting a small set of informative peaks.

Classical *peak picking* applies smoothing, normalisation and signal-to-noise thresholds to each spectrum on its own [Gibb and Strimmer 2012; Deininger et al. 2011]. It depends on manually set thresholds, can discard weak but informative ions [Lieb et al. 2020] and ignores *where* in the tissue a signal occurs. *Peak learning* instead treats selection as an unsupervised representation-learning problem: a model is fitted to the spectra of a section and m/z bins are ranked by how the model uses them [Li et al. 2022; Abdelmoula et al. 2021]. msiPL trains a variational autoencoder (VAE) on individual spectra and reads peaks from its weights [Abdelmoula et al. 2021]. S^{3}PL adds spatial information by convolving over small tissue patches and reconstructing the whole neighbourhood, and reports large gains over msiPL [Weigand et al. 2026a].

S^{3}PL changes the architecture, the objective, the normalisation and the selection rule at the same time, so it is not clear which ingredient produces its gain. This raises two questions. Can cheaper local context, such as the mean spectrum of a pixel's neighbours, give a lightweight VAE a similar benefit? And once a model has been fitted, which part of the pipeline actually determines peak quality: the representation, the segmentation built on it or the method used to rank peaks? We study these questions with *Spatial-msiPL*, an msiPL-style VAE that can encode a spectrum together with its neighbourhood. Its latent space is segmented with a Gaussian mixture model (GMM), and peaks are ranked either by Integrated Gradients (IG) [Sundararajan et al. 2017] through the model or by correlating each ion image with the GMM posterior maps. Specifically, we ask:

RQ1 Does the VAE pipeline select better peaks than msiPL when both select the same number of bins?

RQ2 Which stage of the pipeline determines peak quality: the attribution method or the segmentation, and does the segmentation need a nonlinear encoder?

RQ3 Where does spatial information help: as input to the encoder, in the segmentation or in the ranking statistic?

RQ1 checks that the pipeline is worth studying; RQ2 and RQ3 locate the source of any gain, which a single headline score against a baseline cannot do. To make these comparisons trustworthy, we also reproduce both baselines and analyse how sensitive S^{3}PL is to the training seed.

The main ranking comparisons used matched bin counts per section and the same fair-scoring evaluator. Their decision rules were written down before the corresponding results existed, and the core model comparisons were repeated over three training seeds. The msiPL and S^{3}PL reproductions were first assessed under their respective native protocols and evaluators; S^{3}PL's in-job mSCF1 is distinct from the fair-scoring result. The main findings are as follows.

- **Ranking.** With the segmentation held fixed, correlating each

ion image with the GMM posterior maps outperformed IG in 140 of 144 paired comparisons across three training seeds. The difference was statistically supported for every dataset and model.

- **Spatial information.** In these tests, neighbourhood input to the encoder gave no consistent, spatially specific gain, while spatially aware ranking statistics were the strongest label-free rankings on GBM and RCC and within 0.01 of the best on CAC.
- **Encoder.** A linear five-component PCA segmentation performed comparably to the VAE, so the nonlinear encoder was not shown to be necessary.
- **Reproducibility.** msiPL reproduced. The published S^{3}PL glioblastoma score lay outside the distribution of ten reproduced seeds, and weight initialisation dominated the seed effect in two of three tested sections; data order also mattered in one GBM section.

The contributions of this report are:

1) a ranking ablation showing that segment-then-correlate outperforms gradient attribution in this pipeline, with paired statistics over three training seeds;
2) controlled tests of spatial context at the encoder, segmentation and ranking stages, with dataset-specific ranking outcomes;
3) a statistical account of the S^{3}PL reproduction gap and its seed sensitivity;
4) a matched-budget evaluation protocol with reproduced baselines and model-free references, released with the code.

Section II reviews MSI peak selection and the methods compared. Section III describes the data, models, evaluation, statistics and computation. Section IV presents the results, section V interprets them and discusses limitations, and section VI concludes.

II. Background and Related Work

A. MSI data and preprocessing

An MSI experiment rasters a tissue section and records a spectrum at each location, giving a data cube with two spatial dimensions and one m/z dimension [Caprioli et al. 1997; Buchberger et al. 2018]. Spectra vary in total intensity between pixels for technical reasons, so they are usually normalised, most often by total ion current (TIC), before comparison [Deininger et al. 2011]. Spectra may be kept as fine m/z bins or reduced to centroided peaks. Because neighbouring bins of one molecular peak are strongly correlated, any evaluation that counts bins can reward several bins of the same peak. This motivates the peak-level scoring used in this report (section III-C). Tools such as Cardinal [Bemis et al. 2015] and M^{2}aia [Cordes et al. 2021] provide standard data handling. For our work, the key consequence is that every method must receive the same input and be scored at the same resolution, or differences in preprocessing masquerade as differences in method.

B. Peak selection

Classical pipelines such as MALDIquant detect local maxima above a noise threshold in each spectrum [Gibb and Strimmer 2012]. Machine-learning peak pickers learn what a peak looks like from annotated spectra [Li et al. 2022]. Both operate spectrum by spectrum and do not use the spatial arrangement of the tissue. Learned representation methods instead rank m/z bins by how a model trained on the whole section uses them. This report compares such learned rankings with each other and with simple model-free rankings, under one shared evaluation.

C. Learned peak selection: msiPL and S^{3}PL

msiPL trains a fully connected VAE [Kingma and Welling 2014] on individual spectra, clusters the low-dimensional latent space and reads peaks from the network weights [Abdelmoula et al. 2021]. Its companion work, massNet, provides the glioblastoma (GBM) dataset used here [Abdelmoula et al. 2022]. S^{3}PL learns a 3D convolution over a spectral patch around each pixel, applies an attention step to the centre spectrum, reconstructs the full normalised patch and selects the bins that receive the most attention across pixels [Weigand et al. 2026a]. S^{3}PL also introduced the mask-based spatial-correlation F1 score (mSCF1) used to judge peak lists against expert annotations, and released the peak budgets and protocols for GBM, colorectal adenocarcinoma (CAC) and renal cell carcinoma (RCC). It reported 0.496, 0.601 and 0.470 mSCF1 on these collections, against 0.403, 0.431 and 0.159 for msiPL. Related representation-learning work in MSI includes deep models of 3D data [Inglese et al. 2017], low-dimensional ion-image embeddings [Guo et al. 2024] and IonMorphNet [Weigand et al. 2026b]. S^{3}PL's result motivates our spatial question. Its convolution, objective, normalisation and selector all differ from msiPL, so it is not clear which ingredient produces the gain, or whether a much smaller amount of spatial context would suffice. A related caution concerns how representations are judged: Mazarura et al. [2025] found that reconstruction error, cluster quality and downstream classification ranked autoencoders, PCA and NMF differently on MSI data, so a better reconstruction need not mean better peaks.

D. Attribution and segment-then-correlate

Integrated Gradients attributes a model output to input features by integrating gradients along a path from a baseline input to the actual input [Sundararajan et al. 2017]. Applied to a GMM posterior computed from a VAE latent space, it ranks m/z bins by how strongly they move a pixel towards a tissue segment, through the full nonlinear encoder. An older and simpler route is to segment the tissue first and then correlate each ion image with the segments [Alexandrov et al. 2010; Alexandrov and Kobarg 2011]. Our pipeline contains both routes on the same segmentation, which allows a direct comparison.

E. Spatial statistics and spatially aware methods

Ion images of biologically meaningful molecules tend to be spatially structured rather than noisy. Moran's I measures spatial autocorrelation [Moran 1950], and spatial autocorrelation and spatial-chaos measures have been used to judge the informativeness of MSI ion images [Alexandrov and Bartels

TABLE I
DATASETS. SECTIONS ARE THE UNITS OF ANALYSIS; THE INDEPENDENT UNITS BEHIND THEM DIFFER BY COLLECTION. CAC SECTIONS HAVE SLIGHTLY DIFFERENT m/z AXES; THE RANGE SHOWN IS FOR 40TOPL. GBM BINS ARE NOT EVENLY SPACED (MEDIAN 0.0004 DA).

	GBM	CAC	RCC
Tissue	mouse PDX brain	human colon	human kidney
Ionisation	MALDI	DESI	DESI
Sections	8	8	8
Independent units	4 PDX lines	1 tumour	8 subjects
m/z bins	85,062	1,481	10,200
m/z range	102–1000	198–1061	150–1000
Mean bin width (Da)	0.011	0.58	0.083
Pixels per section	2,071–4,524	3,243–4,761	972–3,564
Mask classes (K)	2	3	3
Accession	ST002045	MTBLS415	CardinalWorkflows

2013; Cassese et al. 2016]. Spatially aware segmentation smooths features using neighbourhood weights before clustering [Alexandrov and Kobarg 2011]. Graph-based models of spatial omics and MSI propagate information between neighbouring locations. These are discussed as future work in section VI. Our study places spatial information at three points (the encoder input, the segmentation and the ranking statistic) and tests each one under the same evaluation.

F. Gap addressed

S^{3}PL shows that a spatial model can select better peaks than msiPL, but not *why*: spatial input, a different representation and a different ranking step all change at once. It is therefore unknown where in a learned pipeline spatial information pays off, and whether cheap neighbourhood context is enough. This report isolates those stages one at a time. Doing so requires every method to be scored on equal terms, so matched peak budgets, model-free references and repeated training runs are the means of the study rather than its subject.

III. METHODOLOGY

A. Data

Three public MSI collections were used (table I and figure 1). They are the collections on which S^{3}PL was published [Weigand et al. 2026a], so both baselines could be reproduced under their own protocols and then compared with our methods under a common one.

GBM is the massNet patient-derived xenograft (PDX) glioblastoma collection, acquired by MALDI on a 9.4 T FT-ICR instrument [Abdelmoula et al. 2022]. It is available from the Metabolomics Workbench (project PR001292, study ST002045, doi:10.21228/M8Q70T). It contains eight sections from four PDX lines, two per line, with pathologist annotations of tumour and normal tissue. Sections from the same line are not independent, so GBM results are also summarised per line. **CAC** is a DESI collection of 52 serial sections from one human colorectal adenocarcinoma [Inglese et al. 2017], available from MetaboLights (MTBLS415). We used the eight sections and three-class masks released with S^{3}PL [Weigand et al. 2026a], so CAC comparisons are within one specimen. Following S^{3}PL, the three classes are tumour, healthy tissue and background; because the expert annotations were not released, these masks are the predictions of a deep-learning model with a reported accuracy of 99.9%, as in the S^{3}PL evaluation. **RCC** is the DESI renal cell carcinoma collection of Dill et al. [2010], distributed with the Bioconductor package CardinalWorkflows [Bemis et al. 2015]. It contains eight runs, each a matched pair of cancer and normal tissue from one subject. Pixels outside the annotated tissue have no label; this unannotated region covers 45–78% of each run. The dataset's own workflow treats it as background, but this interpretation was not independently verified. We kept it as a third class in the primary analysis and report a cancer/normal-only sensitivity analysis (section IV-G). Measured coordinates and masks were audited for alignment in every section.

Four sections (GBM108_positive, GBM22_2, 160TopL and 40TopL) were used for development and tuning, and the method was frozen before any of the comparisons reported here were run. RCC was not used during development and serves as a held-out confirmation collection.

B. Pipeline

Figure 2 summarises the pipeline. For each section, a model is fitted to that section's spectra without labels, peaks are ranked from the fitted model, and the selected peaks are then scored against that section's expert masks. The design is therefore unsupervised and transductive. The masks never enter model fitting or ranking. They do, however, set two configuration values, the number of segments and the peak budget, both disclosed below.

Input. Let $\mathbf{s}_i \in \mathbb{R}^D_{\geq 0}$ be the spectrum at pixel i over D bins. Spectra are normalised by total ion current (TIC), $\mathbf{x}_i = \mathbf{s}_i / \sum_d s_{id}$. For the context arms, let $\mathcal{N}(i)$ be the measured pixels among the eight neighbours of i in a 3×3 window. The uniform context is

$$\mathbf{c}_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j, \tag{1}$$

and the encoder input is $\mathbf{u}_i = \mathbf{x}_i$ (centre-only) or $\mathbf{u}_i = [\mathbf{x}_i; \mathbf{c}_i]$ (uniform). Pixels at tissue edges use only their measured neighbours, so no unmeasured location is padded.

Model. Following msiPL [Abdelmoula et al. 2021], the VAE has one hidden layer of 512 units with batch normalisation and a 5-dimensional latent space [Kingma and Welling 2014]:

$$\mathbf{h}_i = \mathrm{ReLU}\big(\mathrm{BN}(W_e \mathbf{u}_i + \mathbf{b}_e)\big), \tag{2}$$

$$\boldsymbol{\mu}_i = W_\mu \mathbf{h}_i, \quad \log \boldsymbol{\sigma}_i^2 = W_\sigma \mathbf{h}_i, \quad \mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\sigma}_i \odot \boldsymbol{\epsilon}, \tag{3}$$

with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$. The decoder mirrors the encoder and outputs $\hat{\mathbf{x}}_i = \mathrm{sigmoid}(W_o\,\mathrm{ReLU}(\mathrm{BN}(W_d \mathbf{z}_i)))$, a reconstruction of the centre spectrum only. With $p_{id} = \hat{x}_{id} / \sum_{d'} \hat{x}_{id'}$, the loss is msiPL's categorical cross-entropy plus the Kullback–Leibler divergence to a standard normal prior:

$$\mathcal{L}_i = -D \sum_d x_{id} \log p_{id} - \tfrac{1}{2} \sum_{l=1}^{5} \big(1 + \log \sigma_{il}^2 - \mu_{il}^2 - \sigma_{il}^2\big). \tag{4}$$

Training uses Adam (learning rate 10^{-3}, batch size 128) for 100 epochs. For GBM the centre-only and uniform models

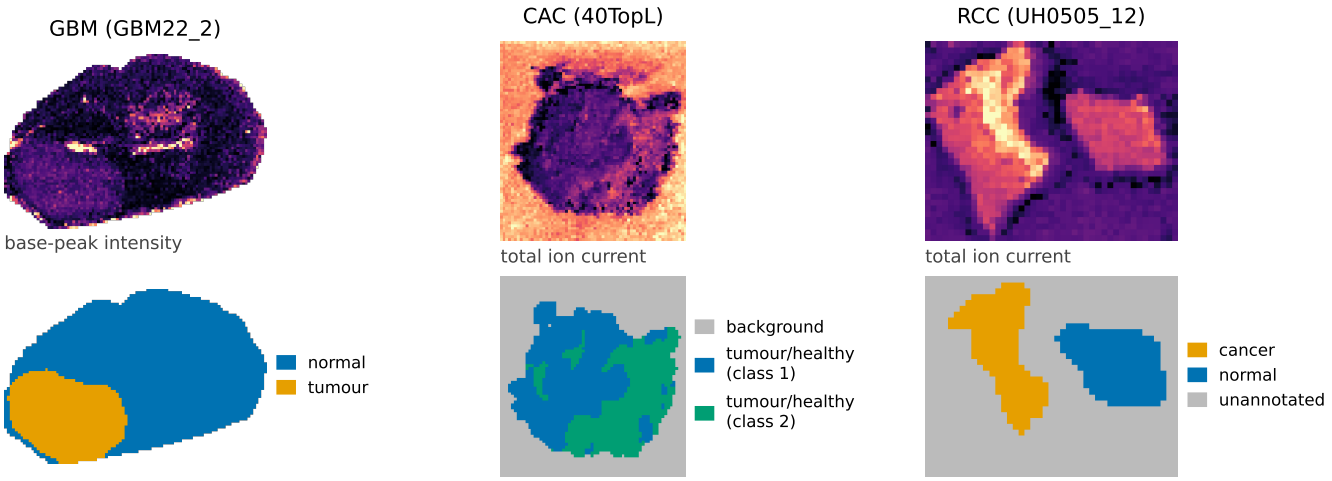


Fig. 1. One representative section per collection (GBM22_2, 40TopL and RCC_UH0505_12). Top: base-peak intensity for GBM, whose spectra are supplied TIC-normalised, and total ion current for CAC and RCC. Bottom: annotation. Unmeasured locations are blank; grey marks the CAC background and the RCC unannotated region. The two CAC tissue classes are tumour and healthy tissue, in an order the dataset does not document.

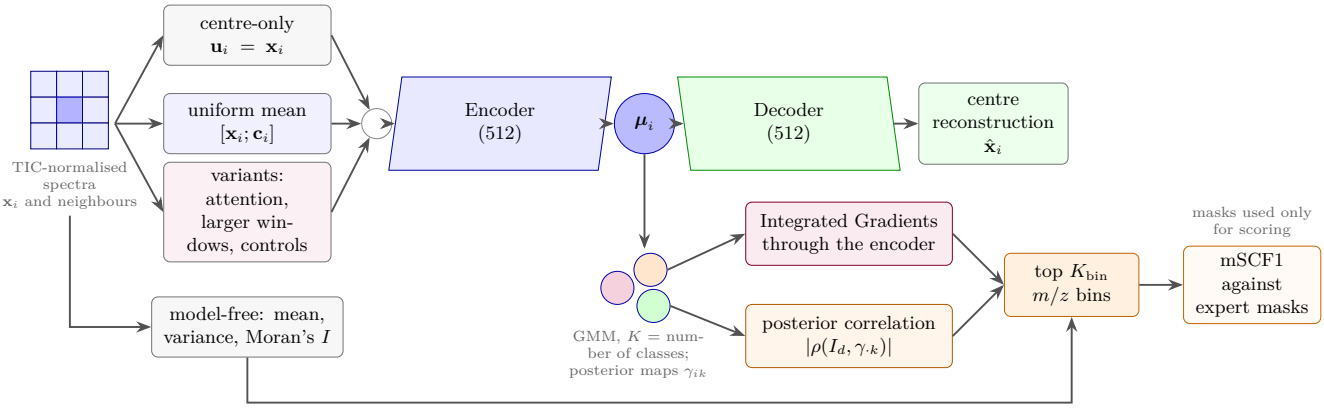


Fig. 2. Pipeline. A VAE is fitted to one section's spectra, with or without neighbourhood context; its latent means are segmented by a GMM; and m/z bins are ranked either by Integrated Gradients through the encoder or by correlation of each ion image with the GMM posterior maps. Model-free rankings use the spectra directly. Expert masks are used only for scoring.

have 87.2M and 130.8M parameters, almost all in the first layer. Further variants replace the uniform mean with learned single-head attention over the neighbours [Vaswani et al. 2017] (about 0.5% more parameters), use 5×5, 7×7 or 9×9 windows, replace the neighbours with zeros or with spectra shuffled from distant pixels, or hide the centre spectrum in half of the training samples so that the neighbours become necessary. Each variant was compared with the uniform arm trained under the same seed.

Segmentation. The latent means $\boldsymbol{\mu}_i$ are standardised and a Gaussian mixture model with full covariance is fitted to them. Its posterior

$$\gamma_{ik} = \frac{\pi_k \, \mathcal{N}(\tilde{\boldsymbol{\mu}}_i \mid \mathbf{m}_k, \Sigma_k)}{\sum_{l=1}^{K} \pi_l \, \mathcal{N}(\tilde{\boldsymbol{\mu}}_i \mid \mathbf{m}_l, \Sigma_l)} \tag{5}$$

gives, for each component k, a posterior map over the section. The number of components K equals the number of annotated classes (table I). This is a class-informed choice; the Bayesian information criterion (BIC) [Schwarz 1978] did not support it (section IV-E). The pipeline is therefore label-free in fitting and ranking but not fully autonomous.

Rankings. Every ranking orders all D bins, and the top K_{bin} bins are scored. The two model-based rankings use the same fitted GMM. For both, a score is computed per component and the per-component orderings are merged round-robin, skipping repeated bins, so that every component contributes equally ("balanced").

- *Integrated Gradients* attributes component k's posterior at pixel i to each input bin along a straight path from a baseline $\mathbf{u}'$, the section's mean input [Sundararajan et al. 2017]:

$$\mathrm{IG}_d(\mathbf{u}_i) = (u_{id} - u'_d) \int_0^1 \frac{\partial \gamma_k\big(\mathbf{u}' + \alpha(\mathbf{u}_i - \mathbf{u}')\big)}{\partial u_d} \, \mathrm{d}\alpha, \tag{6}$$

approximated with 64 trapezoidal steps for 12 sampled pixels assigned to each component. Attributions to a bin's centre and context inputs are combined, and bins are scored by mean absolute attribution. Attributions must satisfy completeness, $\sum_d \mathrm{IG}_d \approx \gamma_k(\mathbf{u}_i) - \gamma_k(\mathbf{u}')$. Where this check failed (4 of

144 runs), the run was recomputed from its saved model with 128 or 256 steps; in four control runs, doubling the steps changed mSCF1 by at most 0.0003.

- *Posterior correlation* scores bin d for component k by $|\rho(I_d, \gamma_{\cdot k})|$, the absolute Pearson correlation over pixels between the bin's ion image I_d and the component's posterior map. It needs no gradients; this is the segment-then-correlate approach [Alexandrov and Kobarg 2011].
- *Model-free rankings* use the spectra directly: mean intensity, variance and global Moran's I [Moran 1950],

$$I_d = \frac{N}{\sum_{i,j} w_{ij}} \frac{\sum_{i,j} w_{ij}(x_{id} - \bar{x}_d)(x_{jd} - \bar{x}_d)}{\sum_i (x_{id} - \bar{x}_d)^2}, \qquad (7)$$

  with $w_{ij} = 1$ for measured Moore neighbours and 0 otherwise.
- *References:* a random ranking, and a supervised oracle that ranks bins by correlation with the expert masks. The oracle is a ceiling only.

Controls on the representation and segmentation. A *PCA control* replaces the VAE latent space with a 5-component PCA while keeping the GMM and posterior correlation, which isolates the nonlinear encoder. A *smoothed segmentation* smooths the latent means with structure-adaptive neighbourhood weights before the GMM [Alexandrov and Kobarg 2011], which places spatial information at the segmentation step instead of the encoder input.

Baselines. Legacy msiPL was reproduced from its released code, with its peak-selection threshold tuned towards the published peak counts as in the original procedure [Abdelmoula et al. 2021]. S^{3}PL was run from the public repository (v1.0.0) with the released settings: 10 epochs, Adam (learning rate 10^{-2}), batch size 16, and patch size and peaks per patch of $p = 3$, $z = 256$ (GBM), $p = 9$, $z = 256$ (CAC) and $p = 5$, $z = 1024$ (RCC) [Weigand et al. 2026a]. The released code normalises each patch by its spatial maximum, whereas the paper describes TIC normalisation, so both were tested (section IV-F).

C. Evaluation

mSCF1. For each class mask and threshold $t \in \{0.3, 0.4, 0.5, 0.6\}$, a reference set R_t of bins is formed from the bins whose ion images correlate with the mask, using the threshold-derived rank cutoff of the released S^{3}PL evaluator [Weigand et al. 2026a]. For a selected set S,

$$\mathrm{mSCF1} = \frac{1}{4} \sum_t \frac{2\,|S \cap R_t|}{|S| + |R_t|}, \qquad (8)$$

with R_t the union of the class reference sets. mSCF1 measures spatial agreement with expert annotation, not chemical identity.

Matched budgets. All methods select the same number of bins, K_{bin}, per section, so that differences reflect which bins are chosen and not how many. For GBM and CAC, K_{bin} is the tuned legacy msiPL peak count; for RCC it is the released S^{3}PL target, which approximately equals the number of bins correlating with the masks at $t = 0.5$. Budgets of $0.5\times$ to $2 \times K_{\text{bin}}$ were checked as a sensitivity analysis. For msiPL, its saved weights were re-thresholded over $\beta = -2.0$ to 2.5 (step 0.01) and, at each budget, the list whose length was closest to the budget was taken; every ranking was then scored at that list's own length. On RCC the longest msiPL list (1,054–1,289 bins in six of the eight runs) was shorter than K_{bin}, so those points use the longest list and are flagged; on GBM, seven of the smallest budgets were below the shortest msiPL list and are flagged likewise.

Peak-level scoring (P1). Neighbouring bins of one molecular peak would otherwise each count as a true positive. P1 therefore collapses bins within one basin of the section's mean spectrum into a single peak group, and partitions were audited and approved before any score was computed. On CAC and RCC, P1 groups are singletons or near-singletons, so P1 acts there as a second budget check rather than independent evidence.

D. Experimental design

Each comparison in section IV follows a rule written down before its outputs existed, in an append-only protocol that also records deviations and execution logs. The rules differ by comparison. In the context comparisons, a method *helps* if its mean gain is at least $+0.02$ mSCF1 and it gains in at least 6 of 8 sections; methods are *equivalent* if their mean difference lies within ± 0.02. Other complete outcomes are *mixed*. The primary IG-versus-correlation rule instead asks whether IG is lower in at least 7 of 8 sections in every run; its predeclared labels and fall-through cases are reported as written. Missing arms remain incomplete, not mixed. The threshold of 0.02 was chosen as the smallest difference we would act on. The two core arms were trained with three seeds on all 24 sections; the variants were screened against the uniform arm with one seed each. Seed results are reported as the mean $\pm$ SD across seeds of the collection means.

E. Statistical analysis

The outcome labels are accompanied by paired non-parametric statistics over sections: the exact two-sided Wilcoxon signed-rank test [Wilcoxon 1945], the Hodges–Lehmann (HL) median paired difference with its exact 95% confidence interval [Hodges and Lehmann 1963], the matched-pairs rank-biserial correlation r and the sign count. Every pair of methods is scored on the same sections, so the data are paired and the signed-rank test is used rather than the unpaired Mann–Whitney test, which would ignore that pairing. Where three seeds exist, seeds are averaged within each section first, so the unit is the section rather than the run.

The 38 rule-labelled comparisons were grouped into five families by question: ranking (6 tests), versus legacy msiPL (4), encoder context and variants (17), segmentation (9) and S^{3}PL (2). Holm correction [Holm 1979] was applied within each family. With $n = 8$ the smallest attainable exact p-value is $2/2^8 = 0.0078$, so Holm significance is attainable only in families of at most six tests, and effect sizes and intervals carry the main weight. A comparison is called *statistically supported* only if its Holm-adjusted $p < 0.05$ *and* its outcome label points the same way. GBM is also summarised per PDX line ($n = 4$), outside the Holm families. Mixed-effects models with random intercepts for section and seed are reported descriptively.

F. Implementation and computation

The pipeline was implemented in Python with PyTorch and scikit-learn [Pedregosa et al. 2011]. Training ran on the University's MSCluster under Slurm, on RTX 3060 and RTX 3090 GPUs (all timed training on RTX 3090), and ranking and scoring ran on CPU nodes. The code, protocol, result summaries and figure scripts are available at https://github.com/2653934/Spatial-msiPL (release `v1.0`, commit `93596d8ed9b99f7d525be0ee96094dd6f64aa97e`).

Caching. The original data loader re-read each batch from HDF5 on shared storage, and a 100-epoch GBM run took 11.3 h. Profiling showed that preparing a batch took 24.6 s, while the forward and backward passes took about 0.02 s. Reading each section into host memory once (about 13 s) reduced the run to 23 min, a 30× speed-up, with identical loss histories and final weights. The cache costs one float32 copy of the section, $N \times D \times 4$ bytes: 0.70 GB for GBM108_positive and at most 1.54 GB for the largest GBM section, with GPU memory unchanged. The likely cause of the slowdown is many small reads over network storage, which was not measured separately.

Reproducibility checks. Before saved runs were reused, same-seed reruns were checked for exact equality of losses, scores and selected bins. Exact equality of rankings depends on the CPU model, because floating-point summation order differs between processors, so each ranking was timed on the same processor type that produced it. Saved result summaries record evaluator and protocol versions, parameters, input hashes and code hashes, while scripts refuse stale or existing outputs; exact reproduction remains hardware- and path-sensitive.

IV. Results

Scores are mSCF1 at the matched budget K_{bin}, at bin level, with differences paired by section. Results that depend on the trained model are given as mean ± SD over three training seeds, and the statistics of section III-E are reported beside the outcome labels.

A. Baseline reproductions

Each baseline was first run under its own protocol and evaluator. msiPL reproduced closely on all three collections: 0.414 against 0.403 published on GBM, 0.402 against 0.431 on CAC and 0.157 against 0.159 on RCC. For S^{3}PL's native in-job mSCF1, the CAC mean was 0.547 ± 0.038 over ten seeds against 0.601 published. The published CAC value lies just outside our seed distribution, subject to published rounding, and the predeclared conclusion is "not explained by seed selection alone". RCC reached 0.41–0.44 over three seeds against 0.47 published, while GBM reached 0.319 ± 0.042 against 0.496 (section IV-F).

B. Ranking by correlation outperforms IG (RQ1, RQ2)

Table II scores every ranking on the same segmentation and budget.

Against legacy msiPL (RQ1). Both VAE–IG arms beat reproduced msiPL in every GBM and CAC section: 0.518 and 0.514 against 0.414 on GBM, and 0.530 and 0.553 against 0.402 on CAC (table II). The difference is statistically supported in all four comparisons (HL $+0.10$ to $+0.15$, $r = 1$, Holm $p = 0.031$). Repeating the comparison with msiPL re-tuned to $0.5\times$–$2 \times K_{\text{bin}}$, IG stayed ahead in all eight GBM sections at every budget. On CAC it stayed ahead from K_{bin} upwards, but at half the budget msiPL was about as good (IG ahead in 5/8 sections), so the predeclared requirement that the advantage hold at every budget was met on GBM but not on CAC. On RCC, where msiPL cannot produce lists as long as K_{bin} in six of eight runs, IG beat msiPL in all eight runs when both were scored at msiPL's own list length (0.37 against 0.16). This compares complete pipelines and does not show which ingredient causes the gain.

IG against posterior correlation (RQ2). Holding each fitted GMM fixed, ranking bins by their correlation with the posterior maps outperformed IG in 140 of 144 comparisons (8 sections × 3 collections × 2 arms × 3 seeds; table III and figure 3). The four exceptions were small, at most $+0.017$. The predeclared rule asked whether IG was below correlation in at least 7 of 8 sections in every training run. It held in 17 of 18 runs and failed in one CAC run, where IG was below in 6 of 8 sections, so the rule's verdict is that the result "does not hold overall". The direction itself never reversed: all 30 collection means of IG minus correlation (18 at bin level and 12 at peak level) were negative.

Statistics. With seeds averaged within section, the difference is statistically supported in all six collection–arm comparisons at bin level (table III). Correlation scored higher in every section (rank-biserial $r = -1$), and every raw p-value is the smallest attainable for $n = 8$, so this is as strong as eight sections can show. The seed changed the size of the gap only slightly: the three seed means of a collection differed by at most 0.05. At peak level (P1) the result is also supported, except for the CAC uniform arm (-0.054, 7/8, Holm $p = 0.078$). The descriptive mixed models agree and assign almost no variance to the seed, and GBM is negative in every PDX line. Sampling 48 or 192 pixels per component for IG instead of 12 raised its score by at most 0.024, far less than the gap.

Answer to RQ1 and RQ2. The pipeline beats legacy msiPL, but the attribution method is not what carries that gain. Given the same segmentation, a direct correlation ranking is better than IG on all three collections. The useful signal is in the label-free segmentation, not in the gradients.

C. Simple rankings and the metric

Among the label-free rankings, the strongest differs by collection (table II): posterior correlation on GBM (0.762, with Moran's I at 0.684), mean intensity on CAC (0.612, with correlation, Moran's I and variance all within 0.02) and Moran's I on RCC (0.631, above correlation at 0.54–0.56). CAC is close to saturation at its budget: all methods, including the oracle, converge near 0.41 at twice the budget, which limits how much CAC differences can reveal. On GBM and RCC, sections whose GMM agrees better with the masks have correlation scores closer to the oracle (Spearman $\rho = -0.90$ and -0.84 against the gap to the oracle). mSCF1 rewards ion

TABLE II

COLLECTION MEAN MSCF1 AT K_{BIN} (BIN LEVEL). IG AND POSTERIOR CORRELATION USE THE SAME FITTED GMM; MODEL-FREE RANKINGS USE THE SPECTRA. BOLD: BEST LABEL-FREE RANKING PER ROW. THE ORACLE USES THE EXPERT MASKS AND IS A CEILING. RCC USES CLASS-INFORMED GMM $K = 3$; 45–78% OF ITS PIXELS ARE UNANNOTATED AND TREATED AS BACKGROUND BY THE CARDINAL WORKFLOW, AN INTERPRETATION NOT INDEPENDENTLY VERIFIED HERE. IG AND POSTERIOR CORRELATION: MEAN ± SD OVER THREE TRAINING SEEDS; THE OTHER RANKINGS DO NOT DEPEND ON THE SEED.

Collection	VAE arm	IG	Posterior corr.	Moran's I	Mean int.	Variance	Legacy msiPL	Random	Oracle
GBM	centre	0.518 ± 0.006	**0.762 ± 0.021**	0.684	0.483	0.497	0.414	0.009	0.809
GBM	uniform	0.514 ± 0.006	**0.757 ± 0.017**	0.684	0.483	0.497	0.414	0.009	0.809
CAC	centre	0.530 ± 0.007	0.608 ± 0.004	0.605	**0.612**	0.593	0.402	0.108	0.654
CAC	uniform	0.553 ± 0.006	0.608 ± 0.002	0.605	**0.612**	0.593	0.402	0.108	0.654
RCC	centre	0.403 ± 0.013	0.543 ± 0.025	**0.631**	0.498	0.508	0.157[a]	0.171	0.756
RCC	uniform	0.421 ± 0.011	0.561 ± 0.019	**0.631**	0.498	0.508	0.157[a]	0.171	0.756

[a] Scored at msiPL's own list length (median 1,158 bins against a median K_{bin} of 1,688), because msiPL cannot reach K_{bin} in six of eight runs. At that length IG scores 0.37 and posterior correlation 0.49.

TABLE III

IG MINUS POSTERIOR CORRELATION (BIN LEVEL). MEAN ± SD ACROSS THREE TRAINING SEEDS OF THE COLLECTION MEANS; IG LOWER: SECTION COMPARISONS IN WHICH IG SCORED LOWER, OVER 8 SECTIONS × 3 SEEDS; HL: HODGES–LEHMANN MEDIAN DIFFERENCE OVER SECTIONS (SEEDS AVERAGED) WITH EXACT 95% CI; HOLM ADJUSTS WITHIN THE SIX-TEST RANKING FAMILY.

	IG − corr.	IG lower	HL [95% CI]	Holm p
GBM centre	−0.244 ± 0.018	24/24	−0.248 [−0.293, −0.186]	0.047
GBM uniform	−0.242 ± 0.011	24/24	−0.241 [−0.311, −0.181]	0.047
CAC centre	−0.077 ± 0.003	23/24	−0.078 [−0.116, −0.040]	0.047
CAC uniform	−0.055 ± 0.007	22/24	−0.046 [−0.080, −0.027]	0.047
RCC centre	−0.141 ± 0.014	23/24	−0.137 [−0.204, −0.073]	0.047
RCC uniform	−0.140 ± 0.026	24/24	−0.139 [−0.210, −0.070]	0.047
Total		140/144		

images that follow the expert regions, so a segmentation that resembles those regions scores well when correlated against; this alignment is part of why correlation does well (section V).

D. Spatial context in the encoder and the segmentation (RQ3)

- **Uniform** 3×3 **context** helps IG on CAC (+0.023 at bin level and +0.027 at peak level over three seeds, gaining in 8/8 sections), although this is not statistically supported (Holm $p = 0.13$), and it gives no gain under correlation ranking (0.000). On GBM and RCC, context and centre-only are equivalent (−0.004, 3/8 and +0.018, 6/8). Across the four GBM PDX lines, context helps some lines and hurts others.
- **Spatial specificity.** If the CAC gain came from correct local placement, real neighbours should beat neighbours shuffled from distant pixels. They tied (0.542 against 0.543). On GBM, zeroed context scored highest (0.459, against 0.446 for real).
- **Larger windows** did not help: 7×7 and 9×9 were 0.035 below 3×3 on GBM (mixed, 2/8) and equivalent on CAC (figure 4).
- **Learned attention** was equivalent to uniform averaging on both collections (+0.011 GBM, −0.006 CAC). With real neighbours its weights stayed almost uniform (normalised entropy 0.98–0.999). With shuffled neighbours it learned selective weights, so the mechanism works; real neighbours are simply near-copies of the centre (median cosine about 0.98). At tissue boundaries, where neighbours differ most, attention did depart further from uniform on the two development sections (median 0.011–0.028 against 0.003–0.011 in the interior), but it favoured neighbours of the same tissue class in only 50–56% of boundary pixels on CAC and 41–50% on GBM, so it did not learn to follow tissue boundaries. Masking the centre during training did not make attention selective, and it lowered GBM scores by 0.039. On two outcome-known development sections, replacing

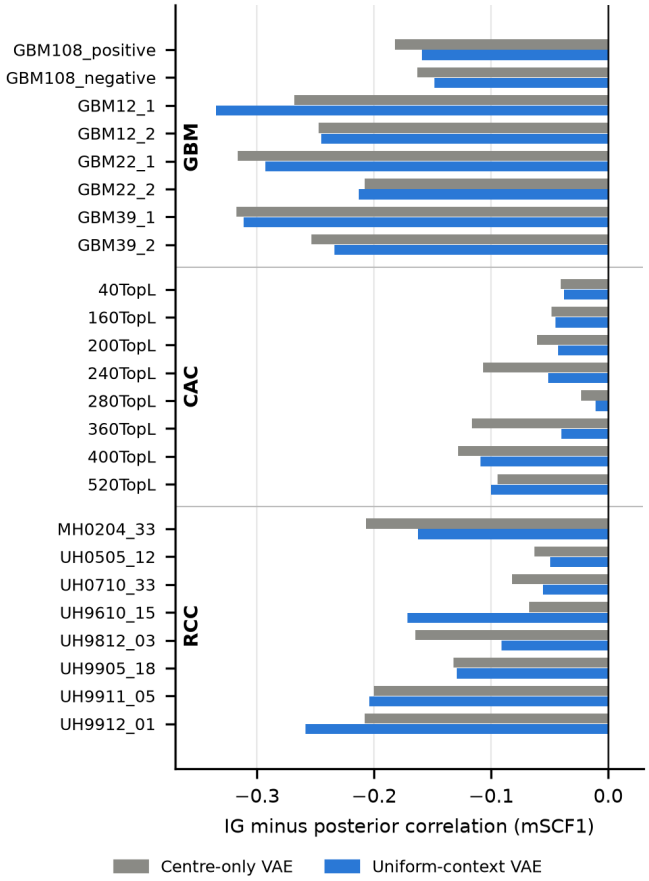


Fig. 3. IG minus posterior correlation (mSCF1) per section for both VAE arms, averaged over three seeds. Every section is negative: correlation with the segmentation outperforms IG on the same segmentation.

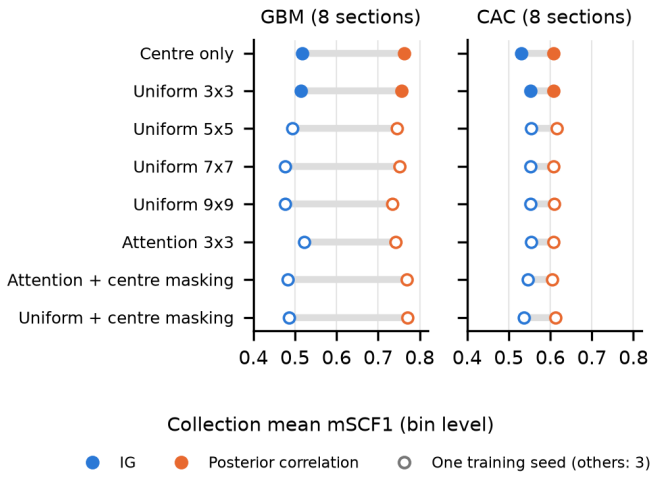


Fig. 4. Neighbourhood variants relative to uniform 3×3 context under IG. No variant gives a consistent, spatially specific gain.

learned attention weights with equal weights at inference slightly raised peak scores ($+0.0099$ and $+0.0038$); this same-checkpoint diagnostic did not retrain a uniform model.

- **Spatially smoothed segmentation** never helped: four comparisons were equivalent and two were mixed, GBM uniform ($+0.029$, driven by two sections) and RCC uniform (-0.021, 4/8).

Answer to RQ3. Neighbourhood information entering the encoder or the segmentation gave no consistent, spatially specific gain in this pipeline. The spatial information that did help sits in the ranking statistic: correlation with spatially coherent segment maps, and Moran's I, which ranks bins by how spatially structured their ion images are.

E. Is the nonlinear encoder needed? (RQ2)

Replacing the VAE latent space with a 5-component PCA, while keeping the GMM and correlation ranking, gave VAE − PCA-5 differences, averaged over three seeds, of $+0.042$ on GBM (higher in 5/8 sections), -0.012 on CAC and -0.010 on RCC (figure 5). The predeclared outcomes were mixed on GBM and CAC and equivalent on RCC, and seed averaging did not change them. VAE segmentations matched the masks slightly better (balanced accuracy 0.868 against 0.839 on GBM) without reliably better peaks. Reconstruction did not separate the two either: PCA-5 had *lower* reconstruction error than the VAE on all eight GBM sections, and slightly higher error on 7/8 CAC sections. The number of GMM components was not validated without labels: BIC selected the largest candidate ($K = 6$) in every tested case, including all 16 RCC runs, with no interior minimum, and a finer segmentation lowered GBM scores by about 0.30.

Longer VAE training reduced the total loss but did not improve mSCF1 monotonically. At 10, 25, 50 and 100 epochs, respectively, the GBM development section gave IG/posterior-correlation scores of $0.509/0.665$, $0.518/0.688$, $0.504/0.671$ and $0.508/0.692$ with centre-only input, while total loss fell from 593,007 through 575,894 and 572,345 to 571,502. With uniform neighbourhood input, its scores were $0.532/0.700$, $0.571/0.704$, $0.564/0.705$ and $0.533/0.705$, with losses 591,713, 575,574, 572,301 and 571,563. On the CAC development section, the centre-only scores were $0.671/0.726$, $0.675/0.727$, $0.661/0.725$ and $0.680/0.727$, with losses 5,813.821, 5,796.127, 5,786.919 and 5,780.477; the uniform-input scores were $0.674/0.733$, $0.665/0.721$, $0.681/0.717$ and $0.683/0.745$, with losses 5,814.905, 5,798.533, 5,788.563 and 5,781.457. Epoch 100 reproduced the production IG and posterior-correlation scores exactly in all four section–arm combinations and both ranking prefixes were identical. The loss decrease was largest by epoch 25 and continued more gradually thereafter, but the scores provide no basis for selecting an epoch.

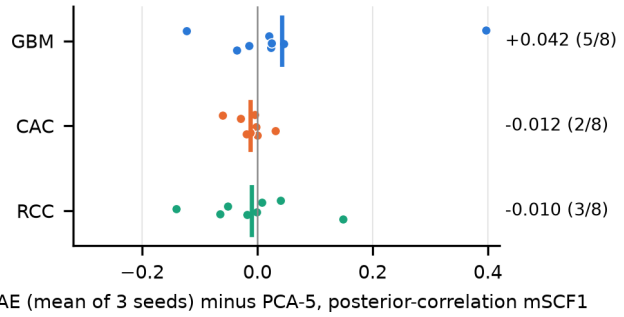


Fig. 5. VAE minus PCA-5 posterior-correlation score per section, with the VAE averaged over three seeds and the same GMM and ranking. Neither segmentation is consistently better.

F. S³PL: neighbour dependence, reproduction and seeds

Neighbour dependence. Retraining S³PL with each patch filled by copies of the centre spectrum collapsed it on CAC from 0.591 to 0.15–0.18 (8/8 sections). Neighbours shuffled from distant pixels recovered most of the score (0.55), and correct placement added about $+0.04$, but inconsistently (mixed, Holm $p = 0.63$). S³PL therefore needs neighbour input under its patch objective, but a consistent benefit from *correctly placed* neighbours was not established. Training for longer made it worse (CAC 0.591 to 0.335 and GBM 0.316 to 0.212 from 10 to 100 epochs; epoch 10 was best in 15 of 16 sections). The released 10-epoch protocol therefore acts as an effective early-stopping choice, but these scores provide no label-free rule for selecting that checkpoint.

Reproduction gap and seeds. Over ten released seeds (figure 6), the published GBM score of 0.496 lies above all ten seed means (maximum 0.374) and above every seed in 6/8 sections; even the best of ten seeds in every section reaches only 0.434. On CAC, the best of ten reaches 0.600, within the published rounding of 0.601. Replacing the released spatial-maximum normalisation with the TIC normalisation described in the paper did not close the GBM gap (0.303 against 0.324). We therefore leave the GBM gap unexplained; untested candidates are section-specific data preparation and hyperparameters tuned on the evaluation sections. The seed effect itself is specific: about half of the score variance lies between sections, almost none in a seed main effect and about 47% in a section $\times$ seed interaction. Seeds switch particular

TABLE IV
MEDIAN RUNTIME PER SECTION. TRAINING ON GPU; RANKING ON CPU (TWO THREADS) GIVEN A TRAINED MODEL, WITH IG INCLUDING THE GMM FIT. WITHIN EACH COLLECTION, ALL RANKINGS WERE TIMED ON THE SAME CPU TYPE. ALL TRAINING TIMES ARE ON AN NVIDIA RTX 3090; RANKING RAN ON AN INTEL CORE I5-14400 (GBM, CAC) AND AN INTEL CORE I9-10940X (RCC).

	GBM	CAC	RCC
VAE training (min)	32–35	1.3–1.4	2.2
S^{3}PL training (min)	13.7	0.9	1.6
Posterior corr. (s)	1.12	0.028	0.11
Moran's I (s)	8.4	0.18	0.75
Mean intensity (s)	0.10	0.003	0.013
IG (s)	770	6.0	372

sections between distinct solutions (GBM108_positive between 0.02 and 0.46). Weight initialisation dominated in two of three selected sections (SD 0.222 against 0.001 from data order on GBM108_positive), while data order also mattered on GBM108_negative. The measured contribution of GPU nondeterminism was limited in the tested repeats.

Summary. msiPL reproduced. S^{3}PL's native results were close on RCC and CAC, subject to the CAC seed-distribution verdict above, but its published GBM value remains unreproduced. Individual sections can switch between distinct solutions, making single-run comparisons unreliable.

G. RCC: held-out confirmation

On the held-out RCC collection, correlation outperformed IG in every run in both arms, and Moran's I was the strongest label-free ranking (0.63). The primary GMM used class-informed $K = 3$. The unannotated class covers 45–78% of pixels; the Cardinal workflow treats it as background (`rcc_nobg`), but we did not verify that interpretation independently. The documentation describes heterogeneous tumour with large normal-looking areas in UH0505_12 and UH9905_18. When only cancer and normal pixels are scored the advantage shrinks: correlation beats IG in 5/8 runs for the centre-only arm (and falls below mean intensity, variance and Moran's I) and in 6/8 runs for the uniform arm. The three-class RCC result is therefore partly driven by separating annotated tissue from the unannotated region; RCC is held-out confirmation of the ranking contrast under these qualifications.

H. Runtime

Once a segmentation exists, correlation takes about a second or less per section, while IG takes seconds to minutes (table IV and figure 7). Every timed ranking with a saved comparator reproduced its ranking exactly. Cached 100-epoch VAE training takes longer than S^{3}PL's 10-epoch protocol but remains practical. Over a complete workflow on CAC section 160TopL (training, selection and scoring, three repeats), the VAE took 122–135 s against 151–174 s for S^{3}PL, with 1.2 against 1.3 GiB of host memory and 237 against 315 MiB of GPU memory. These are measurements under each method's own protocol, not an equal-work comparison.

V. DISCUSSION

A. Why correlation outperforms IG here

Three explanations fit the ranking result.

1) **Different questions.** IG asks which bins would change a few sampled pixels' component probabilities. Correlation asks which ion images follow a component across the whole section, which is closer to what mSCF1 rewards.
2) **Sampling.** IG averages over 12 sampled pixels per component, while correlation uses every pixel. More pixels improved IG only slightly ($\leq +0.024$) on the two sections tested, so sampling explains little of a gap of 0.05–0.25, but other IG settings (baselines, path counts, signed attributions) remain untested.
3) **Metric alignment.** A segmentation that resembles the masks produces posterior maps that a mask-based metric naturally rewards.

Correlation is therefore best read as an ablation finding about this pipeline: the label-free segmentation carries the useful signal, and gradient attribution through a dense VAE does not add to it. We do not claim that IG is generally inferior, nor that segment-then-correlate, which is established practice [Alexandrov and Kobarg 2011], is a new method.

B. What remains of the spatial hypothesis

The original hypothesis was that neighbourhood input would improve a VAE's representation and therefore its peaks. In a dense VAE that averages or attends over neighbours, no intervention gave a consistent gain beyond what nonlocal (shuffled) context also provides, and smoothing the segmentation did not help either. The attention experiments explain why. At 3×3 scale, adjacent spectra are near-copies of the centre, so there is little to select between. Spatially aware ranking statistics were strongest on GBM and RCC and within 0.01 of the best on CAC. S^{3}PL collapses without neighbour input under its patch objective, although correctly placed neighbours added only an inconsistent $+0.04$ over shuffled ones. In this pipeline, then, spatial information paid off when it was used to *rank* peaks, not when it was fed to the encoder: spatially aware ranking rather than spatially aware encoding is what the evidence supports, within the tested models and datasets.

C. Reconstruction is not ranking quality

PCA-5 reconstructed GBM better than the VAE yet ranked peaks about equally well, while the VAE segmented slightly better. Reconstruction error rewards fitting all intensity variation, including intense ions unrelated to tissue classes, whereas peak quality here depends on how well a low-dimensional segmentation separates the classes. This agrees with Mazarura et al. [2025].

On two outcome-known sections, label-trained probes using within-section pixel folds recovered expert-class information from frozen latents even when the unsupervised GMM partitioned them differently; the probes are diagnostics, not label-free validation. On one section, a tied-covariance GMM agreed better with the masks but had worse BIC, so selecting it for mask agreement would be label-guided.

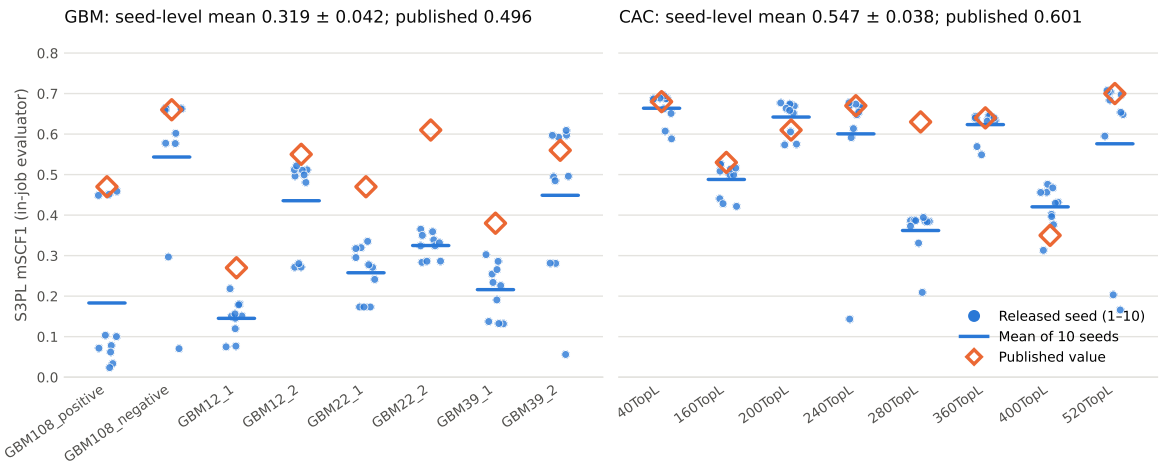


Fig. 6. S^{3}PL mSCF1 per section over ten released seeds (dots), seed mean (bar) and published value (diamond), using S^{3}PL's own evaluator. On GBM the published value is above every seed in 6/8 sections; on CAC it lies just outside, within the published rounding.

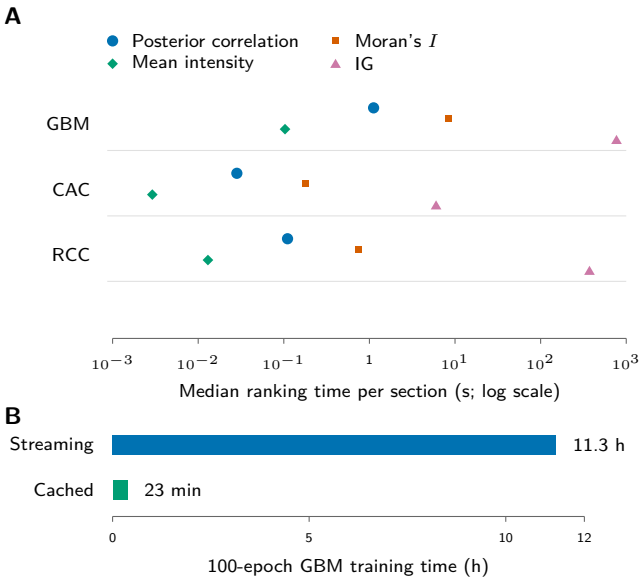


Fig. 7. Measured runtime. (A) Median ranking time per section across eight sections in each collection (log scale). Posterior correlation includes GMM posterior calculation; IG includes GMM fitting and faithfulness checks. GBM and CAC use the same CPU type within each collection; RCC uses shared CPU nodes. (B) The audited 100-epoch GBM uniform-context VAE run before and after caching on an RTX 3090. These timings compare an implementation change, not methods.

D. S^{3}PL instability and the reproduction gap

The S^{3}PL seed analysis shows that comparisons with a single S^{3}PL run can be misleading. Specific sections switch between distinct solutions depending on the initial weights. The released preprocessing appears to create this instability, since TIC normalisation removes it while also removing the highest-scoring outcomes. Neither seeds nor normalisation explain the published GBM score. We therefore report the gap as unresolved, and we do not select favourable seeds or checkpoints for any baseline.

E. Label-free operation and cost

The pipeline fits and ranks without labels, but two settings still come from annotation: the number of segments and the peak budget. BIC could not replace the first, so the configuration remains class-informed. On cost, the dense VAE used comparable time and less memory than S^{3}PL on the section measured, and once a segmentation exists, correlation ranking takes about a second where IG takes minutes. The lightweight model is therefore practical, but it did not gain from its spatial input, which was the reason for adding it.

The conclusions are bounded to the tested models and datasets: we do not claim that spatial context never helps, that IG is generally inferior or that a nonlinear encoder is never needed.

F. Limitations

Each collection has eight sections, drawn from four PDX lines (GBM), one tumour (CAC) and eight subjects (RCC), so the statistics describe these sections rather than a wider population, and effect sizes carry most of the weight. Two settings come from the annotations, the number of segments and the peak budget, so the pipeline is label-free in fitting and ranking but not yet autonomous. mSCF1 measures agreement with tissue masks rather than molecular identity, which favours methods that produce mask-like maps; the RCC masks include a large unannotated region, and the CAC masks are model predictions. Finally, we tested fixed-window neighbourhoods in a dense VAE. Graph-based and image-level spatial models, transfer between sections and other attribution settings remain open (section VI).

VI. Conclusion

This report tested whether local spatial context helps a lightweight VAE select informative MSI peaks, and which part of a learned pipeline determines peak quality. Under matched budgets and predeclared rules, on three public collections, we reached the following conclusions.

- Ranking peaks by correlation with a label-free GMM segmentation outperformed gradient attribution through the VAE on every dataset. This was statistically supported and stable across training seeds.
- Neighbourhood input to the encoder and smoothing of the segmentation gave no consistent, spatially specific gain. Spatially aware ranking statistics were strongest on GBM and RCC and within 0.01 of the best on CAC.
- A linear PCA segmentation performed comparably to the VAE, and reconstruction error did not track peak quality.
- msiPL reproduced. S^3PL's published GBM score did not. Weight initialisation dominated seed sensitivity in two of three selected sections, while data order also mattered in one.

The contribution is a controlled evaluation, not a new peak picker. It shows that, in a learned MSI pipeline, the useful spatial signal lies in how peaks are ranked against a segmentation rather than in neighbourhood input to the encoder, and how strongly such conclusions depend on matched budgets, model-free references and seed replication.

Future work should first:

- validate a label-free choice of the number of segments, which BIC could not provide;
- test a predeclared combination of segmentation correlation and Moran's I on new data;
- evaluate spatial models that operate on graphs or whole ion images rather than fixed windows;
- test whether a model fitted on one section transfers to another.

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools [ ]

[X] I **did** make use of generative AI tools for the following (tick all that apply):

1.	Idea Generation (research problem/design, hypothesis)	X
2.	Sourcing Related Work (summarising, identifying sources)	X
3.	Methods and Experiment Design (experiment setup, model tuning)	X
4.	Data Analysis (presentation, coding, interpretation)	X
5.	Theoretical Development (theorem proving, conceptual analysis)	
6.	Code Development (generating algorithms, writing scripts)	X
7.	Presentation (rendering graphics, formatting)	X
8.	Editing (grammar, readability)	X
9.	Writing (text generation, document structuring)	X
10.	Citation Formatting (structuring, organising)	X

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**
ChatGPT (OpenAI)	Early supervision and planning: experiment ideas, decision rules and reviewing results.
Claude (Anthropic)	Supervision: planning experiments, decision rules, reviewing results against saved outputs, drafting and editing the report.
Claude Code (Anthropic)	Writing, testing and running analysis code and cluster jobs.
OpenAI Codex	Code review, commits and report editing and checking.

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2653934

Candidate signature:



Date: 30 October 2026